

Default Simulation Parameters

for “Joint Task Offloading and UAV Trajectory Optimization for Space–Air–Ground Integrated Mobile Edge Computing Networks”

Default values for the experimental evaluation (Section 5); sweeps vary only task size and MUE/UAV/LEO counts.

Parameter	Symbol	Value
Scenario and MUEs		
<i>Service area</i>	–	$1000 \times 1000 \text{ m}^2$
<i>Number of MUEs (default)</i>	N	200
<i>MUE positions</i>	–	uniform over the service area
<i>Task input size</i>	A_i	$\in [5, 20] \text{ MB}$
<i>Computing density</i>	ν	1000 cycles/bit
<i>MUE CPU capability</i>	f_i^{local}	$\in [2.0, 3.0] \text{ GHz}$
<i>Transmit power</i>	p_i	$\in [0.5, 1.5] \text{ W}$
<i>Energy per CPU cycle</i>	γ_{e_i}	10^{-9} J/cycle
<i>Cost weights</i>	λ_t, λ_e	0.5, 0.5
UAV tier (default $M = 10$)		
<i>Altitude</i>	Z_{u_j}	80 m
<i>Service radius</i>	R_{uav}	400 m
<i>Computation capacity</i>	F_u^{max}	10 GHz per UAV
<i>Max. movement budget</i>	D_{max}	65 m (from initial position)
BS tier (default $X = 2$)		
<i>Computation capacity</i>	F_b^{max}	100 GHz per BS
<i>Service radius</i>	R_{bs}	300 m
LEO tier (default $K = 4$)		
<i>Orbit altitude</i>	h	800 km
<i>Earth radius</i>	R	6371 km
<i>Orbital speed</i>	V_s	7.5 km/s
<i>Minimum elevation angle</i>	α	20°
<i>Computation capacity</i>	F_s^{max}	30 GHz per satellite
Channels and PHY		
<i>Orthogonal channels per node</i>	–	3
<i>Channel bandwidth</i>	W	5 MHz
<i>Background noise</i>	σ^2	10^{-9} (normalized)
<i>G2A/G2G carrier frequency</i>	f_c	2 GHz
<i>LoS probability parameters</i>	a, b	9.61, 0.16
<i>Excess path loss</i>	η_{los}, η_{nlos}	1 dB, 20 dB
<i>G2S carrier frequency</i>	–	20 GHz (Ka band)
<i>User/satellite antenna gain</i>	G_t, G_r	35 dBi, 45 dBi
<i>Atmospheric loss</i>	L_{atm}	2.5 dB
<i>LEO shadow fading std.</i>	σ_s	3 dB
FDTO		
<i>Finite-difference perturbation</i>	δ	5 m
<i>Candidate step sizes</i>	$\mathcal{S}$	$\{5, 10, 20, 40, 65\} \text{ m}$
<i>Best-response re-evaluation radius</i>	–	$3R_{uav}$
<i>Cost-evaluation radius</i>	–	$2R_{uav}$
Statistical protocol		
<i>Independent runs per measurement</i>	R	10 (seeds 2025–2034)
Stopping rules and baselines		
<i>Outer-loop horizon</i>	T_{max}	20
<i>Early stop (i)</i>	–	no UAV movement accepted in a round
<i>Early stop (ii)</i>	–	$\leq \lfloor M/5 \rfloor$ (min 1) moves; improvement $< 10^{-8}$ for 3 rounds
<i>BRG stopping criterion</i>	–	no improving update in a full sweep
<i>Re-convergence sweep cap</i>	–	$\max(N/3, 50)$ sweeps per outer round
<i>Baseline iteration caps</i>	–	BCD+SCA 15×10 ; Stackelberg 30×5 ; Greedy/CCPM single-pass